

SECONDARY SCIENCE EXAMINATION- 2016

(ANNUAL)

SCIENCE

[TIME: 2 HOURS 30 minutes]

F. M. 80

**Instructions for Candidates:**

1. Candidates are required to give their answers in their own words as far as possible.
2. Figures in the right hand margin indicates full marks.
3. While answering the questions Candidate should adhere to the word limit as far as possible.
4. 15 minutes of extra time has been allotted for the candidates to read the questions carefully.

**Section A – PHYSICS**

1. **What is the resistance of human body?**

Ans: The resistance of human body taken between two hands or between the legs ranges 500 mA to 500 kilo $\Omega$ .

2. **what is the minimum distance of clear vision ?**

Ans: The minimum distance of clear vision is 25 cm.

3. **what is LPG?**

Ans: LPG is Liquefied Petroleum gas.

4. **What is SI unit of electrical charge ?**

Ans: SI unit of electrical charge is coulomb.

5. **What is SI unit of power of lens ?**

Ans: SI unit of power of lens is Diopter.

6. **Write Ohm's Law?**

Ans: According to Ohm's law the electric current flowing through a conductor is directly proportional to the potential difference applied across its end, providing the physical condition such as temperature remains unchanged.

**7. What is an electrical fuse? From which alloy it is made of?**

Ans: Fuse is used as a safety device in household circuits and is based on the heating effect of current. It is connected in series with the mains supply. A fuse consists of an alloy of lead and tin which has appropriate melting point.

**8. What is differentiate between concave mirror and convex mirror?**

Ans: Spherical mirror with inward curved reflecting surface is called a concave mirror. A beam of light generally converges after reflection from such surface hence it is called convergent mirror.

Spherical Mirrors with outward reflecting surface is called convex mirror. A beam of light gently diverges after reflection from the surface and it is called divergent mirror.

**9. Why red colour is used in railway signal?**

Ans: Red colour has highest wavelength of all the colors and it is scattered the least by air particles so it is able to travel the longest distance. This is the reason why it is used in railway signals.

**10. Write for Ohm's law and prove it.**

Ans: According to Ohm's Law

$$V \propto I \quad \text{at constant temperature}$$

$$\text{Or } V = IR$$

$$\text{Or } I = V/R$$

Where R is the constant of proportionality called resistance of conductor at a given temperature.

From the above formula it is clear that current is inversely proportional to resistance.

**OR**

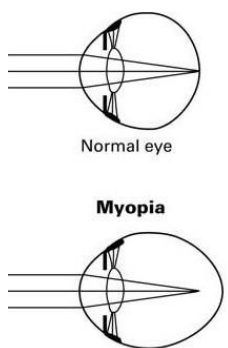
**What is eye defect? How many types of it? Explain anyone with labeled diagram.**

Ans: The defect due to which a person cannot see the object distinctly and comfortably are called defects of vision. The main defects of vision are

- Myopia for short sightedness.
- Hypermetropia or long sightedness.
- Presbyopia.
- Astigmatism.
- colour blindness

Myopia or short sightedness-

In this defect a person can see nearby objects clearly but cannot see distant objects clearly the image is formed in front of retina and not on retina.



## CHEMISTRY

### 1. How rancidity occurred in food material?

Ans: It is a process of slow oxidation of oil and fat present in food materials resulting in the change of smell and taste in them.

### 2. The properties of element are .....of their atomic number.

Ans: The properties of element are periodic function of their atomic number.

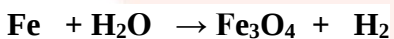
### 3. How many allotropes of carbon are there?

Ans: Different allotropes of carbon are crystalline form for example Diamond graphite, micro crystalline form or amorphous form for example coal, lamp black and charcoal

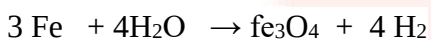
### 4. Write the molecular formula of washing soda.

Ans:  $\text{Na}_2\text{CO}_3$  is the molecular formula of washing soda

### 5. Balance the equation.



Ans:



## Short answer questions

### 6. What is Homologous series?

2

Ans: A series of similarly constituted compounds in which the members present as same functional groups and similar chemical properties and any two successive members in a

particular series differ in their molecular formula but a  $\text{CH}_2$  - unit is called homologous series. For example  $\text{CH}_4$ ,  $\text{C}_2\text{H}_6$ ,  $\text{C}_3\text{H}_8$ ,  $\text{C}_4\text{H}_{10}$  are members of alkane family.

**7. What is sonorous?**

2

**Ans:** The metals that produce a sound on striking a hard surface are said to be sonorous. Bells are made up of metal based on this property.

**8. What are hydrocarbons? How many types are there?**

3

**Ans:** The organic compounds containing only one carbon and hydrogen are called hydrocarbons for example  $\text{CH}_4$ ,  $\text{C}_2\text{H}_6$  and  $\text{C}_2\text{H}_2$ . These are the simplest organic compounds and are regarded as parent organic compounds. The major source of hydrocarbon is petroleum. Hydrocarbons are classified into saturated hydrocarbons and unsaturated hydrocarbons.

**9. Mention the use of the alloy with their components-**

2

- a) Steel                      b) brass                      c) bronze

**Ans: a) Steel-** Steel is an alloy of iron and chromium. Chromium produces a thin layer of oxide on the surface of steel which prevents further corrosion of the surface. Steel is made in various forms including plates, bus seats and tubing for use in industrial and domestic settings.

b) Brass- Brass is an alloy of Copper and Tin. It is not a good conductor of electricity. Brass is used in instruments, coins and money, buried fixtures such as door knobs and bolts.

c) Bronze-

**Long answer questions:**

**10. What is the reason for the excessive number of organic compounds in nature? Explain.**

**Ans:** The main reason for the huge number of carbon compounds are following –

- Catenation - The property of self-linking of elements mainly C-atoms through covalent bonds to form long straight or branched chains and rings of different size is called catenation. Carbon shows maximum catenation in the periodic table due to its small size and high strength of C-C bonds.
- Carbon has 4 electrons in the outermost shell. Its valency is 4 so it is capable of bonding or sharing with four other carbon atoms or with atoms of some other monovalent elements like hydrogen, halogen etc.

- Due to its small size carbon has strong tendency to form multiple bonds by sharing more than one electron pair with its own atoms or with the atoms of elements like oxygen nitrogen sulphur etc.

OR

Match the column (i) with column (ii) .

| COLUMN (i)                                           | COLUMN (ii)    |
|------------------------------------------------------|----------------|
| i) Change of blue litmus into red                    | a) Neutral     |
| ii) Sources of Milk                                  | b) Acidic      |
| iii) Phenolphthalein changes in acidic solution      | c) Lactic Acid |
| iv) Methyl orange changes in basic/alkaline solution | d) colorless   |
| v) pH of a solution                                  | e) Yellow      |

Ans:

|                                                      |                |
|------------------------------------------------------|----------------|
| i) Change of blue litmus into red                    | a) Neutral     |
| ii) Sources of Milk                                  | b) Acidic      |
| iii) Phenolphthalein changes in acidic solution      | c) Lactic Acid |
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## BIOLOGY

### 1. What is the unit of Living Organism?

Ans: The cell is the basic unit of living organism. All living organisms are made up of cells and all functions basic taking place inside the body of organisms are performed by cell. Cells combine to form tissues which further combine to form organ . Organ combined to form organ system which further combine to form organism. So cell is a basic structural unit of all unicellular and multicellular organism.

### 2. How many Chambers are there in human heart?

Ans: There are four chambers in human heart. Two auricles and two ventricles.

### 3. In Dengue which components lack in blood?

Ans: Blood platelets lack in Dengue fever.

### 4. Write full form of ATP.

Ans: Adenosine Tri-phosphate.

### 5. How does exchange of gases takes place in plants?

Ans: Exchange of gases in root of a plant takes place by the process of diffusion from the air present in between soil particle. The gaseous exchange in stem takes place through small pores in stem called lenticel in woody plants. In herbaceous plants stomata on the stem are responsible for the respiratory process. In leaves the respiration takes place by diffusion of oxygen through stomata present in the surface of leaf. Stomata are the aerating pores present on the epidermis of leaf.

## 6 Write four characteristics of Living Organism?

Ans: Characteristics of living things are following-

- Respiration- All living things require energy. The energy is released during the breakdown of energy rich compound by process of respiration.
- Movement- All living things show the movement for what reason may be food shelter or to avoid unfavorable condition. In animals movement of parts is more obvious. They move from place to place. The movement occurs in plant also within the whole structure of plant.
- Growth- All living organism have capacity of growth. Increases in mass and increase in number of individual are twin characteristics of growth.
- Reproduction- Living organism shows the process of reproduction to ensure the survival of species because the lifespan of organism is limited. The resulting offspring have same general characteristics of their parents this process is called reproduction.

## 7. Write four names of organelles of a cell.

Ans: Cell membrane, Nucleus, Mitochondria, Endoplasmic Reticulum.

## 8. Why are we resemble our parents?

Ans: We resemble our parents because characters are transmitted from one generation to generation or from parents to offspring. This is called heredity. DNA copying occurs and some of the characters are inherited to offspring from parents during the process of sexual reproduction.

## 9. How many types of reproduction are there?

Ans: When offspring is produced by single parent without the involvement of gamete, formation the reproduction is asexual. When two parents of opposite sex participate in the reproduction process and also involves the fusion of male and female gamete it is called a sexual reproduction.

### 11. Explain the process of reproduction in flowering plants?

Ans: Vegetative propagation is used in plants as a mean of asexual reproduction. The unit of vegetative propagation such as runner, rhizome, sucker, Tuber, offset, bulb are all capable of giving rise to new offspring. These structures are called vegetative propagules. The formation of this structure does not involve parent, the process is called asexual reproduction.

Androecium consisting of stamen represents the male reproductive structure and gynoecium consisting of pistil represents the female reproductive structure. Pollen grains are present in the anther they represent the male gametophyte generation. The pistil has three parts- stigma, style and ovary. Ovules are present in the ovary. Pollination is the mechanism of transfer of pollen grains from anther to stigma. On reaching stigma pollen tube develops and discharge male gamete. Fertilization occurs and zygote is formed. After fertilization ovary develops into fruit and ovule develop into seeds.

### PART- B

20 MARKS

### PHYSICS

1. The device to generate an electric current is called -----.

- a) Generator                      b) Galvanometer  
b) Ammeter                      d) None of these

Ans: Generator

2. S I unit of energy is

- a) Calorie      b) Joule  
b)      d) none Of These

Ans: Calorie

1. Which of the following means water?

- a) Petro                      b) Turbo  
c) Nitro                      d) Hydro

Ans: Hydro

2. Speed of light in vacuum is

- a)  $3 \times 10^8$  m/s                      b)  $3 \times 10^8$  cm/s  
c)  $3 \times 10^8$  Km/s                      d) None of these

Ans:  $3 \times 10^8$  m/s

5. What is determined by the law of reflection?

- a) Angle i = angle r      b) Angle r = angle i  
c) Angle of Deviation      d) None of these

Ans: Angle i = angle r

6. The number of main focus in the lens is

- a) Two      b) One  
c) Three      d) None of These

Ans: Two

7. When temperature rises the resistance of conductor is

- a) Increases      b) Decreases  
c) Increases Or decreases      d) None of These

Ans: Increases

## CHEMISTRY

1) Which type of reaction is this disintegration of vegetables to become compost?

- a) Reduction      b) endothermic  
b) Exothermic      d) Combination

Ans: Exothermic

2) Oxidation is a process in which

- a) Addition of oxygen      b) Disintegration of hydrogen  
c) Loss of electron      d) All of these

Ans: All of these

3) During the process of electrolysis electron releases on

- a) On anode      b) On Cathode  
c) On both of them      d) none of these



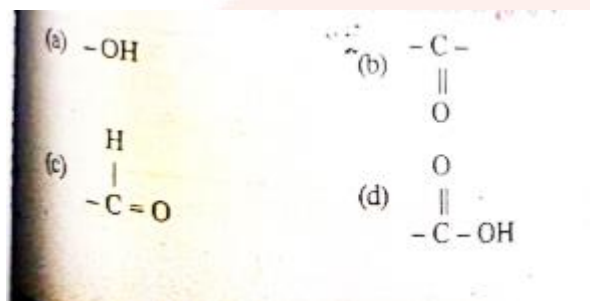
Ans: Cathode

4) Which is the most reactive metal in reactivity series?

- a) Al                      b) Na  
c) Hg                      d) Cu

Ans: Na

5) Reactive group of aldehyde family is



Ans: (C)

6) The general formula for alkyne family is

- a)  $\text{C}_n\text{H}_{2n}$     b)  $\text{C}_n\text{H}_{2n+2}$     c)  $\text{C}_n\text{H}_{2n-2}$     d)  $\text{C}_n\text{H}_n$

Ans:  $\text{C}_n\text{H}_{2n-2}$

7) One PM (picometer) is equal to

- a)  $10^{-12} \text{ m}$                       b)  $10^{-10} \text{ m}$   
c)  $10^{12} \text{ m}$                       d)  $10^{-6} \text{ m}$ .

Ans:  $10^{-12} \text{ m}$

## BIOLOGY

**1. Which is the male gamete of the following-**

- a) Sperm b) Ovary
- c) Vaginad) Fallopian Tube

Ans: Sperm

**2) Gibberlin is**

- a)Hormone b) Fat
- c) Enzyme d) Carbohydrate

Ans: Hormone.

**3) Phloem is responsible for**

- a) Water transportation b) Food transportation
- c) Amino acid transportation d) Carriage of Oxygen

Ans: Water transportation

**4) Foliage of leaves is caused by**

- a) Auxin b) Gibberllin
- c) Cytokinin d) Abcsisic acid

Ans: Auxin

**5) The empty space between two nerve cells is called**

- a) Dendrite b) Synapse
- c) Axon d) Impulse

Ans: Synapse

**6) Asexual reproduction is done by**

- a) Amoeba b) Yeast
- c) Plasmodium d) Malarial Parasite.

Ans: Amoeba